

Multifile project building:

1.c 2.c 3.c 4.c
↓ ↓
l.asm

cl /Ehsc /Fe:app.exe 1.c 2.c 3.c 4.c
gcc -o app 1.c 2.c 3.c 4.c
cl
g++

C / C++ → ASM 1.c

global_num = 50 ; 12.c

char

→ movb \$ 50, global_num

→ movb \$ 50, '1000' M(1000)

M[1001:1001]

short

→ movw \$ 50, global_num M[1000:1005]

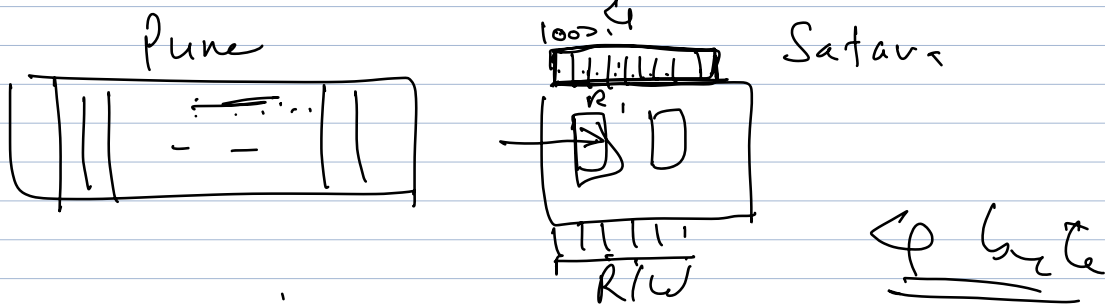
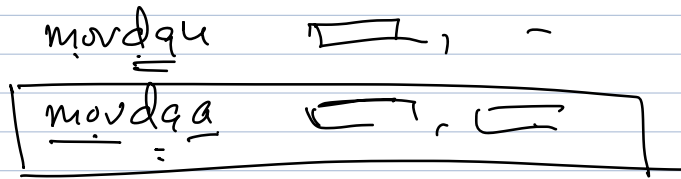
→ movw \$ 50, '1000' M[1001:1007]

int

→ movl \$ 50, global_num

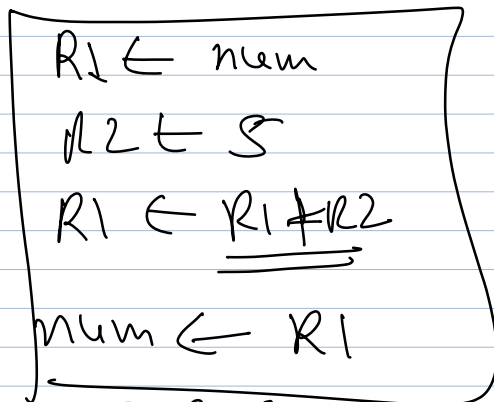
→ movl \$ 50, '1000'

movq



int num;
- M[1000] M[1002:1005]

num = num + 5



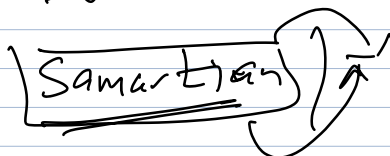
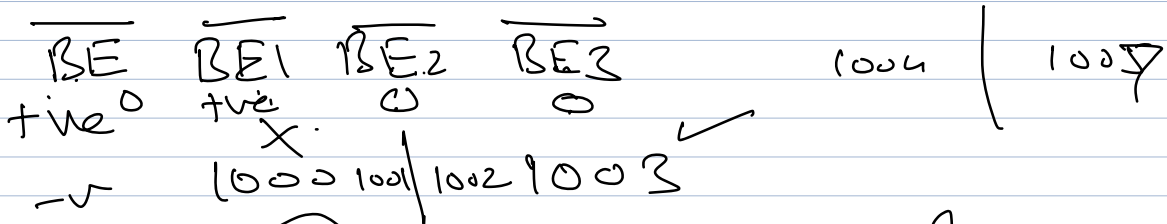
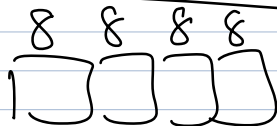
M[1000]

1000: 1003

(1002:1003)

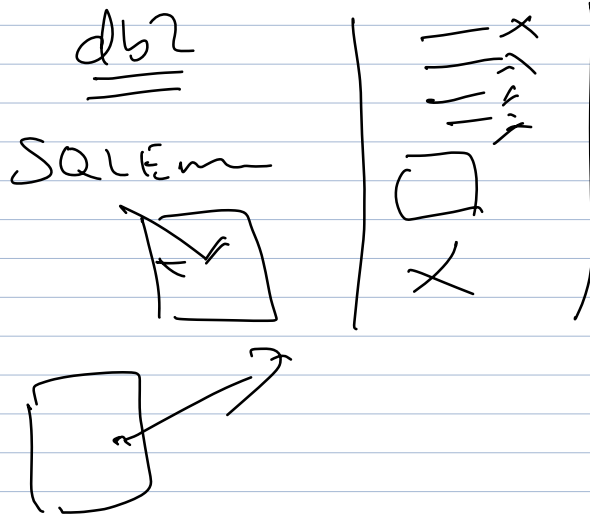
M[1004]: 1007

1004:1005



xmmu

xmm7



1.c

```
int global_num = 100;
1000
```

Data Definition
Statement

type declaration
simple declaration

2.c

```
extern int global_num;
2000

void test_function()
{
    global_num = 50;
}
}
```

2.c

extern int global_num = 100;

Don't do this!!!

1.C	2.C
extern int x;	<u>static int</u> x=10;
extern void	static void test(void);
test(void);	int* get_addr()
	{ return (&x);

extern static int x; ~~X~~ ~~X~~ ~~X~~

extern static void test(void); ~~X~~

type var_name;

int float num;

extern
static . type

[storage class] type var_name [=];

↓

Mr/Mrs. _____

M ~~10~~ 12